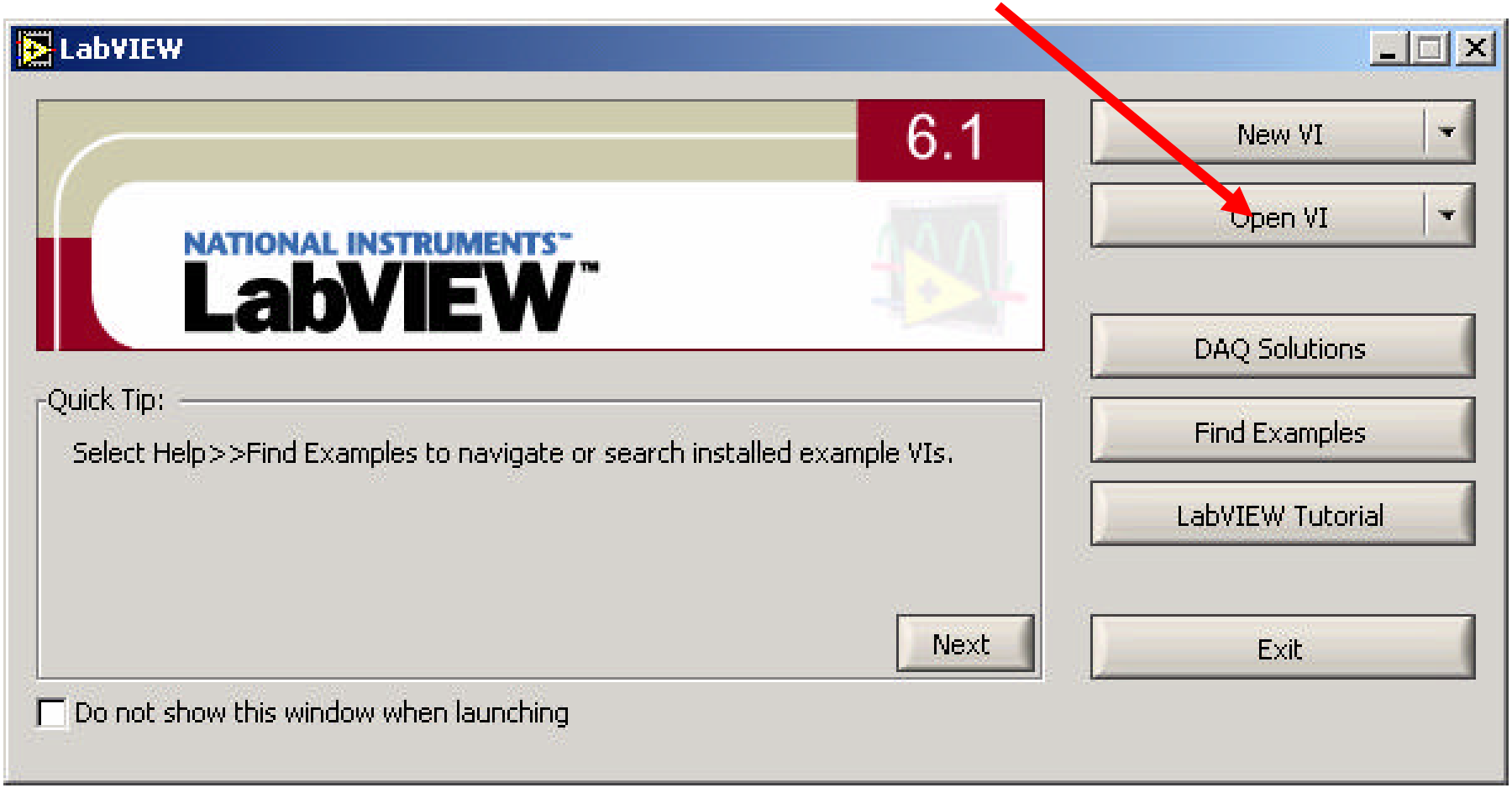
LabVIEW Tutorial

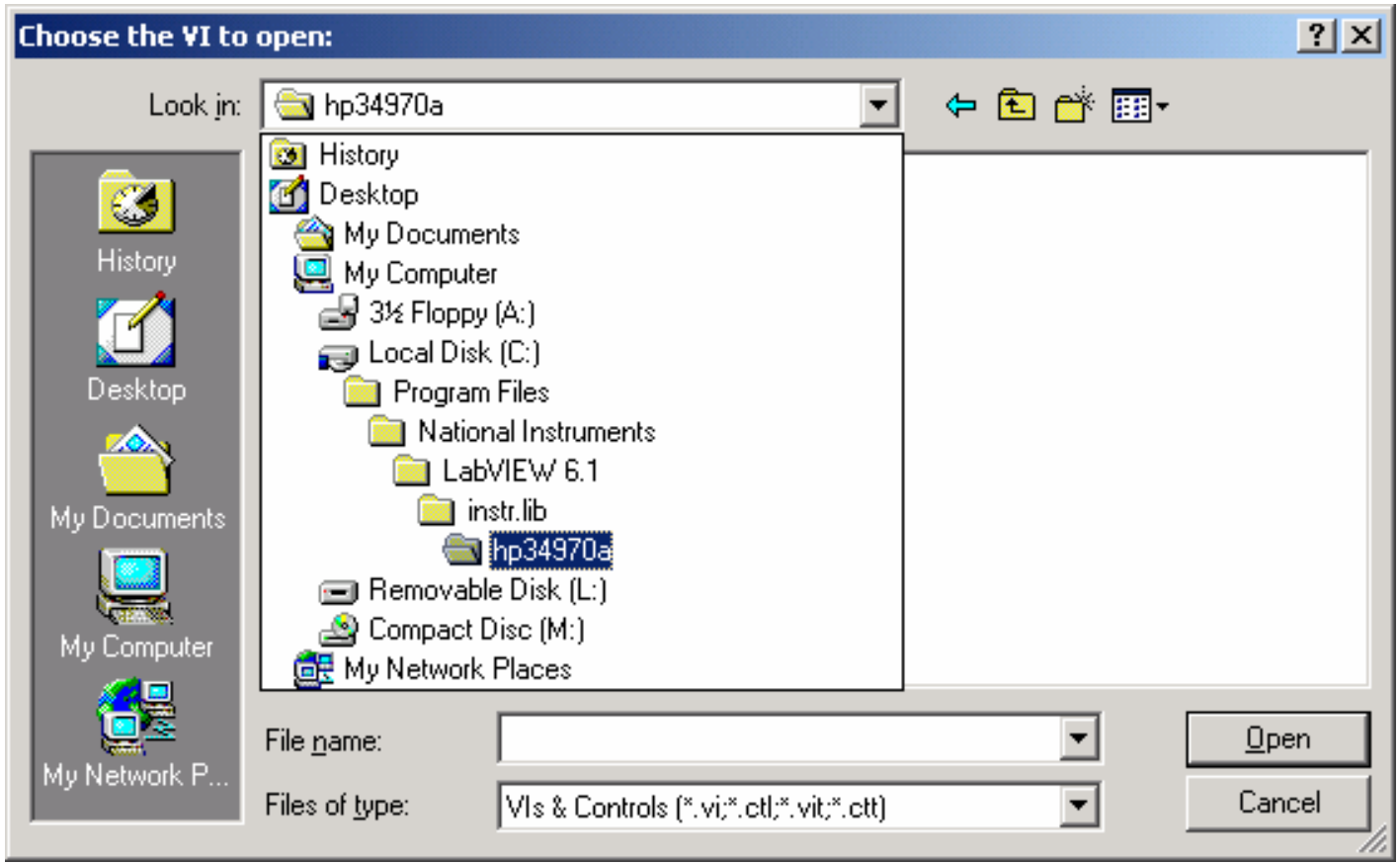
Data Acquisition with Agilent 34970A via RS-232

Part 2

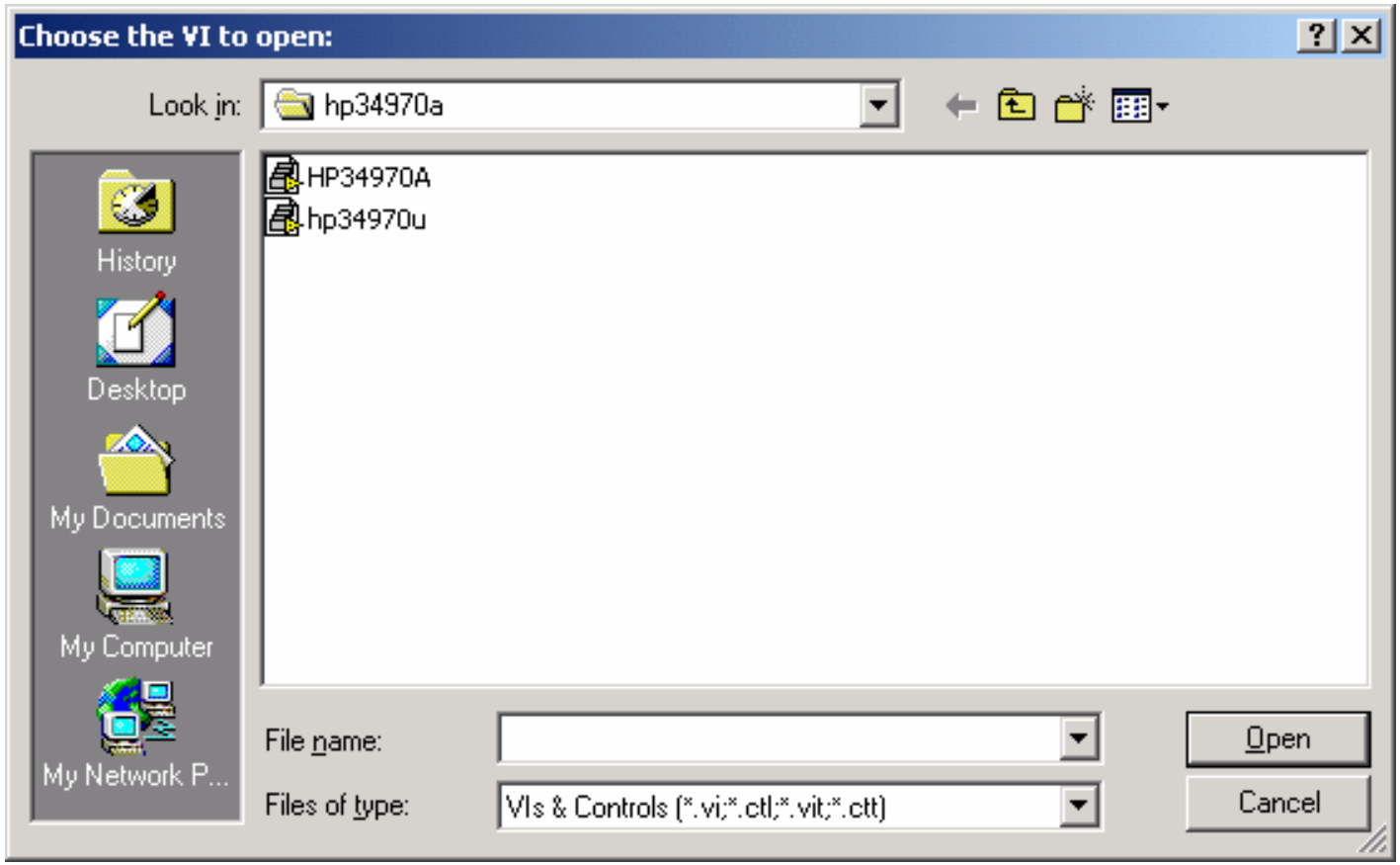
- Please ***read*** and follow the instructions in this tutorial ***carefully***. If you think you've messed something up, the easiest way to fix it for what we're doing today is to start over. **DO NOT** save any changes you may have made to the built-in VI's we're using (HP34970A Scanner Demo.vi). Then just start over… Good Luck!

Since we're using an existing VI…

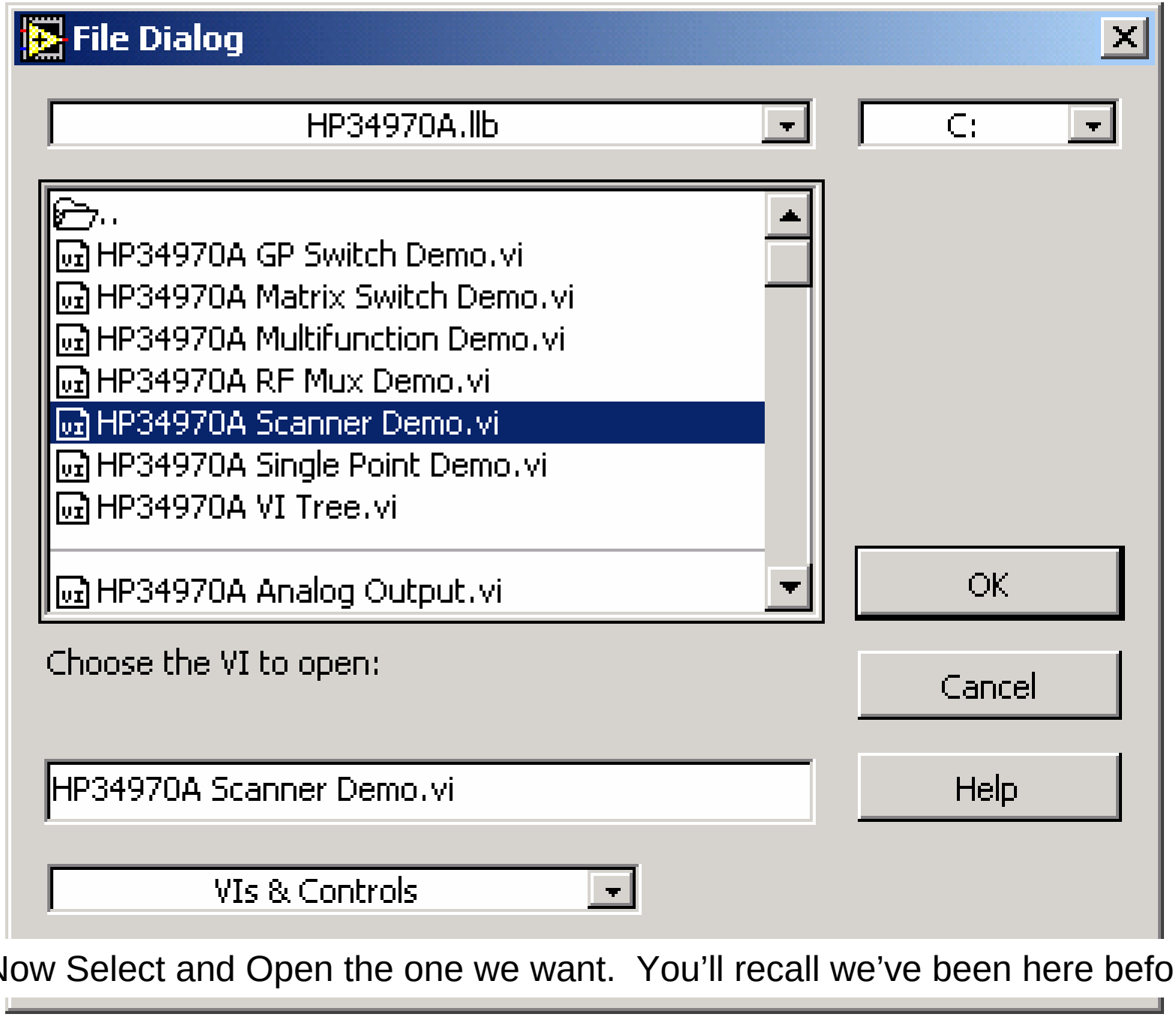




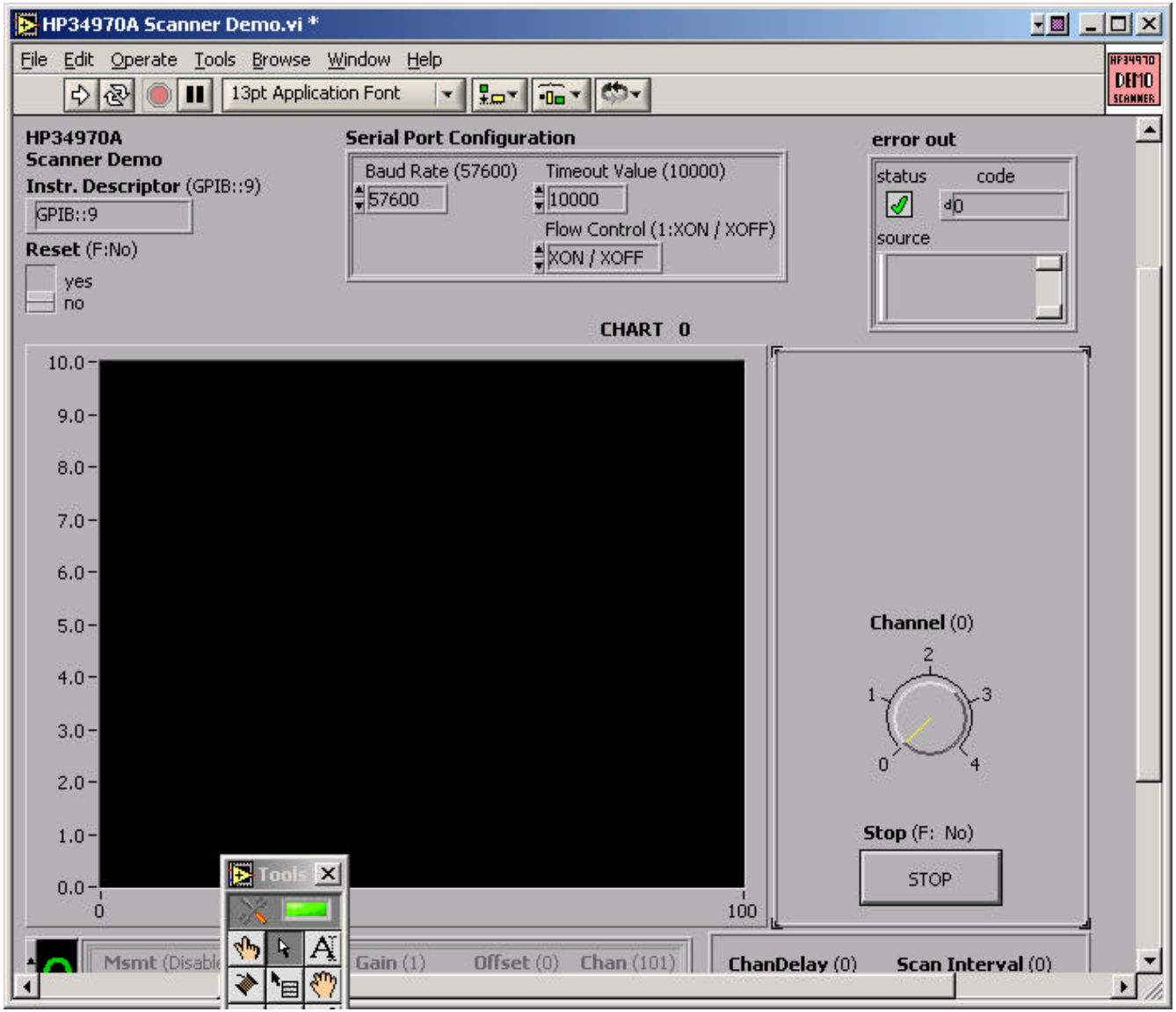
You'll need to surf through the directory structure to find the Agilent 34970A driver files



You should see something like this… Select hp34970a and Open it…

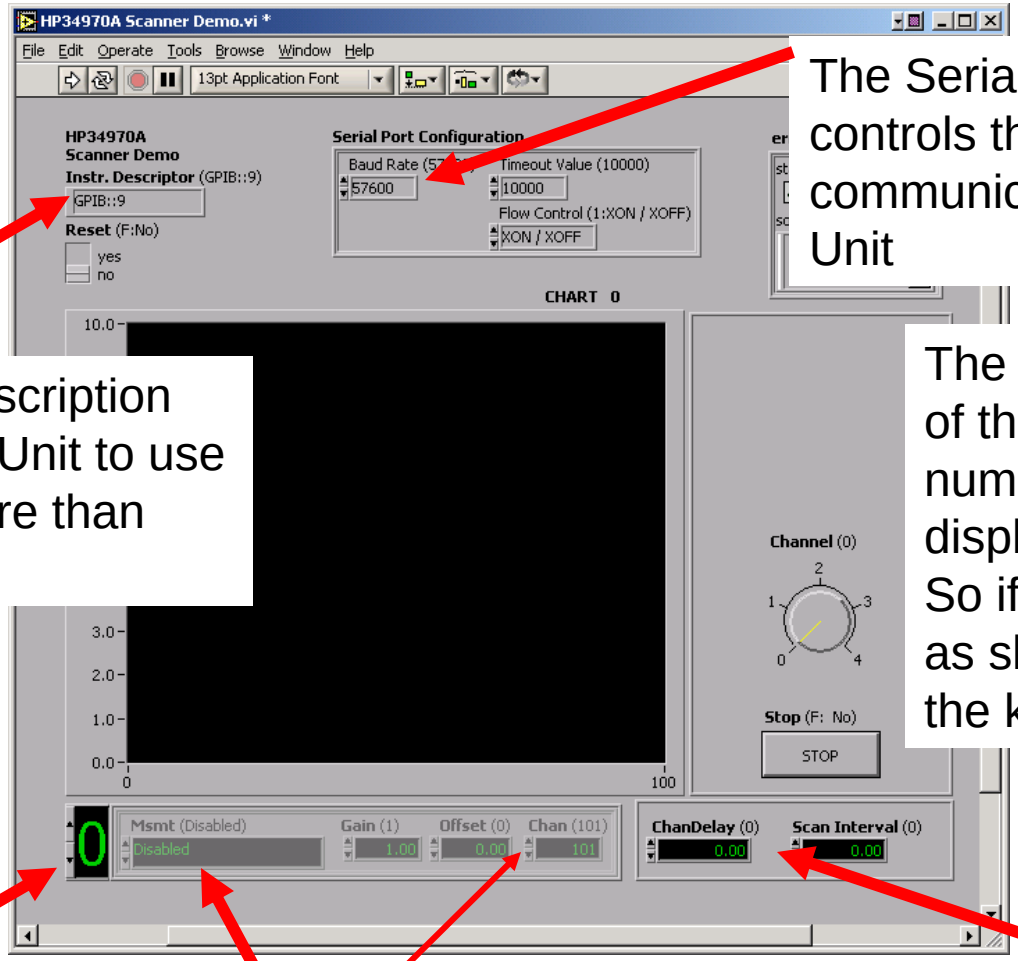


Now Select and Open the one we want. You'll recall we've been here before…



You should see something like this. Now let's look at what we need to do to get it to run properly…

- Why don’t you use the Context Help to see if you agree with my descriptive labeling of the parts in the next slide…



The Serial Port configuration controls the settings for communication with the DAQ Unit

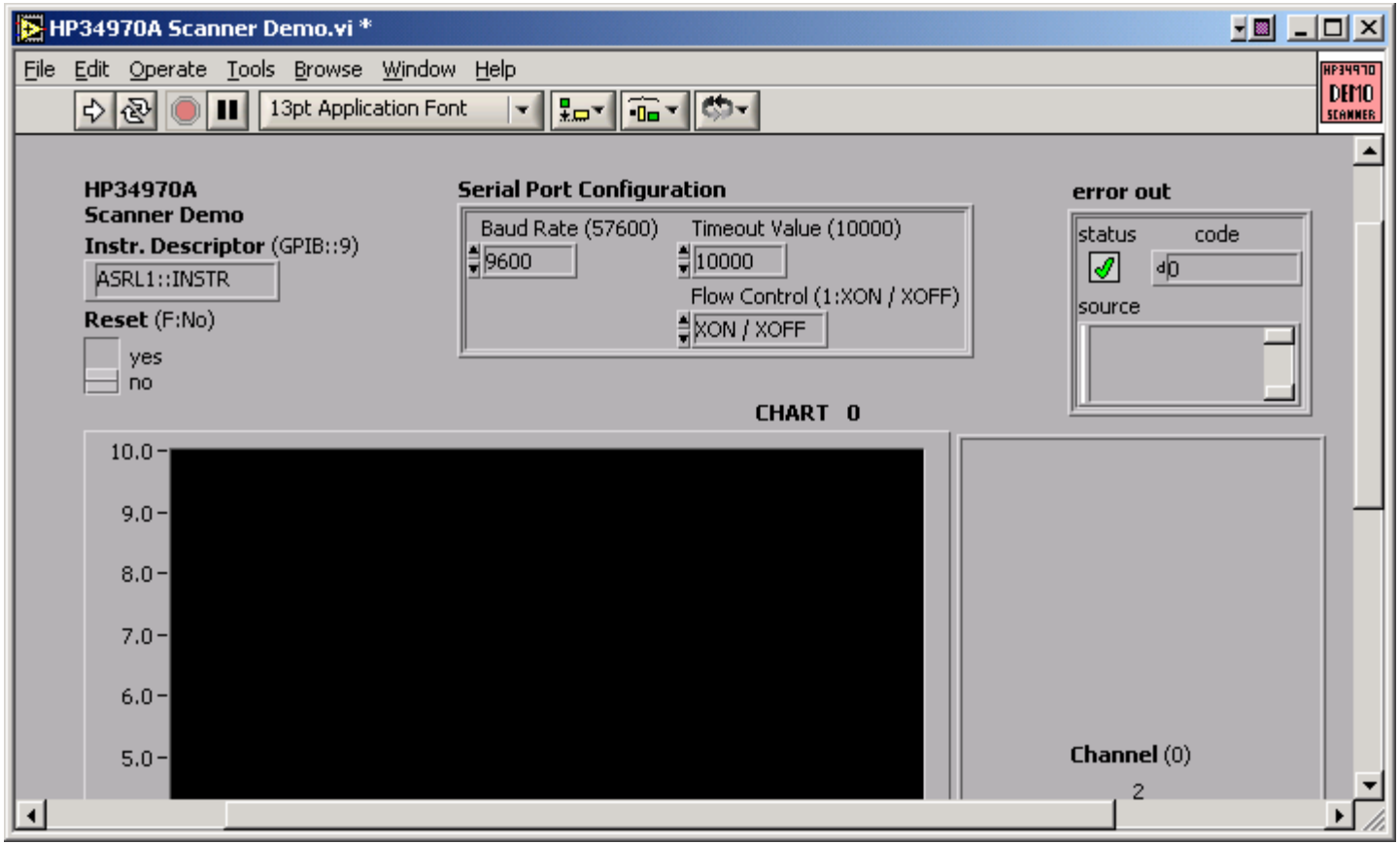
The Instrument Description identifies the DAQ Unit to use (there could be more than one on our PC)

The knob controls which of the Panel signals (big number…) is being displayed on the Chart. So if you're measuring 0 as shown here, you want the knob to be on 0.

Finally, there are timing controls for the measurements…

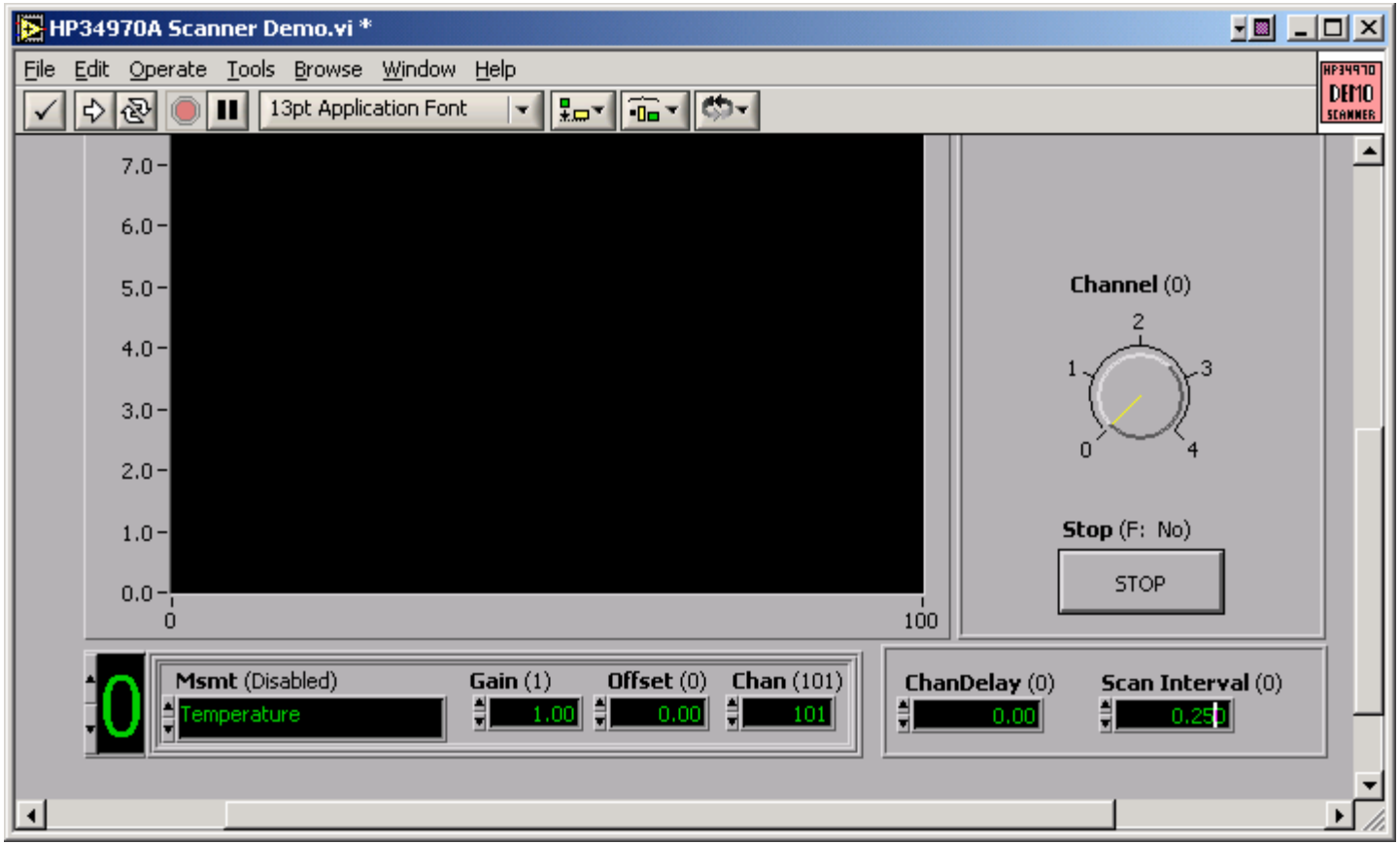
The big number is the Panel's identifier for the channels on your DAQ unit… Note that 0 is representing Channel 1

Also note that the Msmt is disabled... We'll use the Finger tool to Enable it



I’ve used the cursor to type in the Instrument Descriptor information.

I’ve used the Finger tool to SCROLL to the correct Baud Rate value!

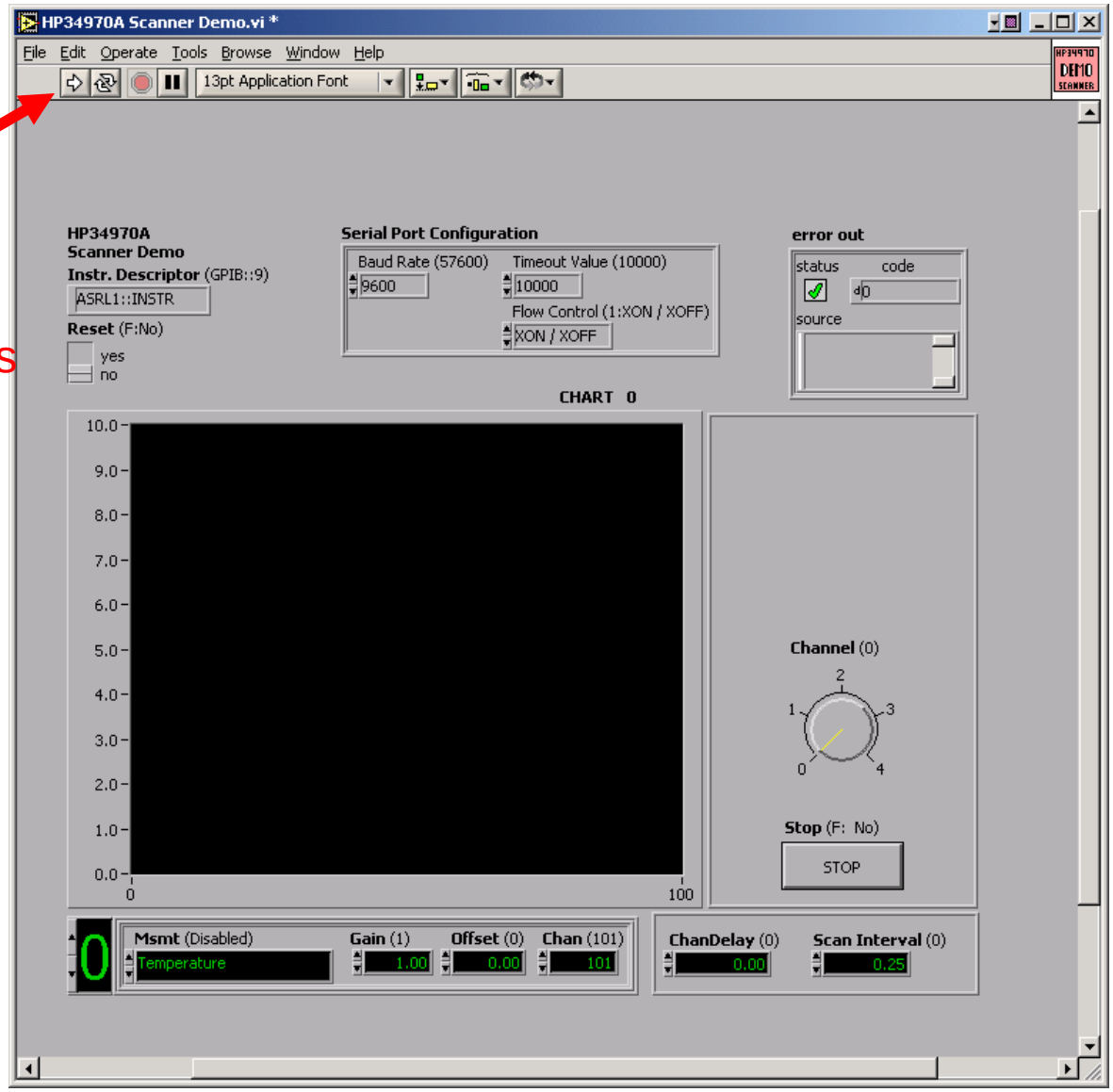


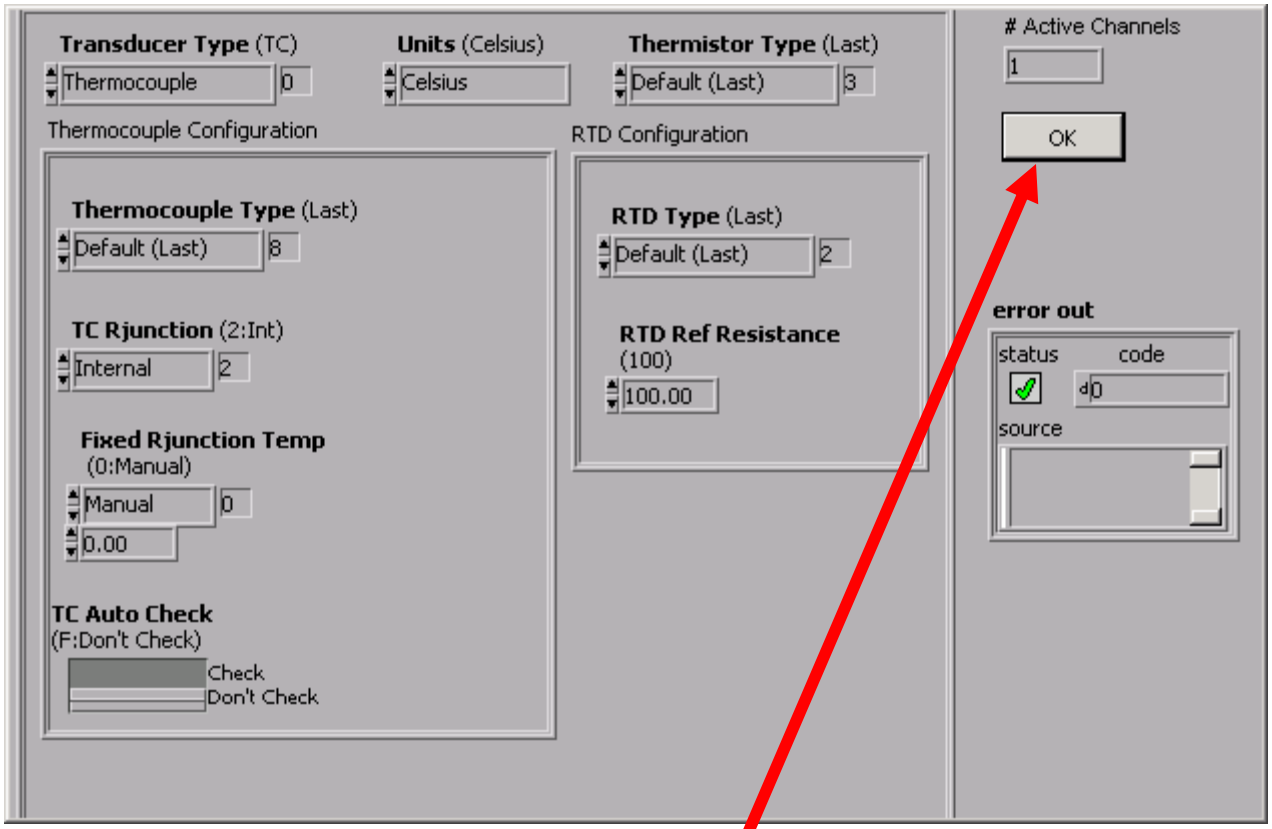
I’ve used the Finger tool to enable the Msmt and select Temperature

I’ve also changed the Scan Interval to ¼ second…

Now I’m ready to run it…

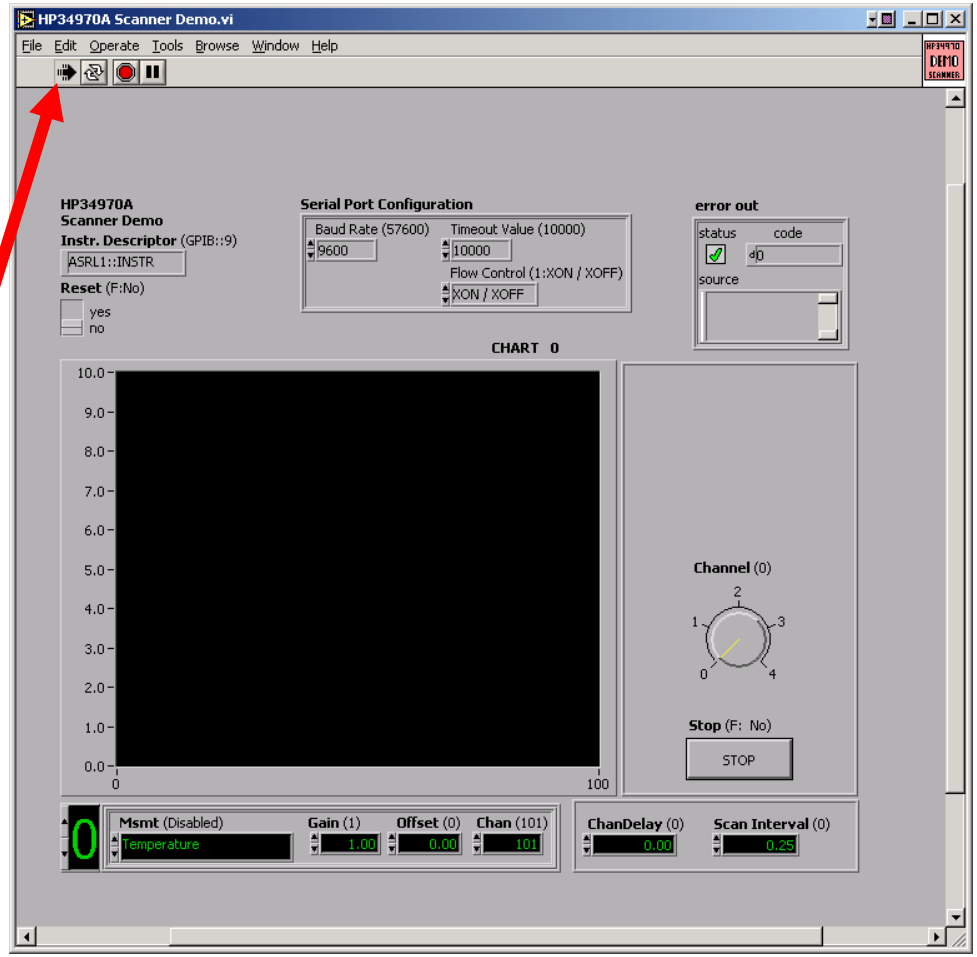
To run the VI, press the run button…





You should see this screen. Don't worry about the Thermocouple settings now; we just want to get some data.

Then press the OK button…



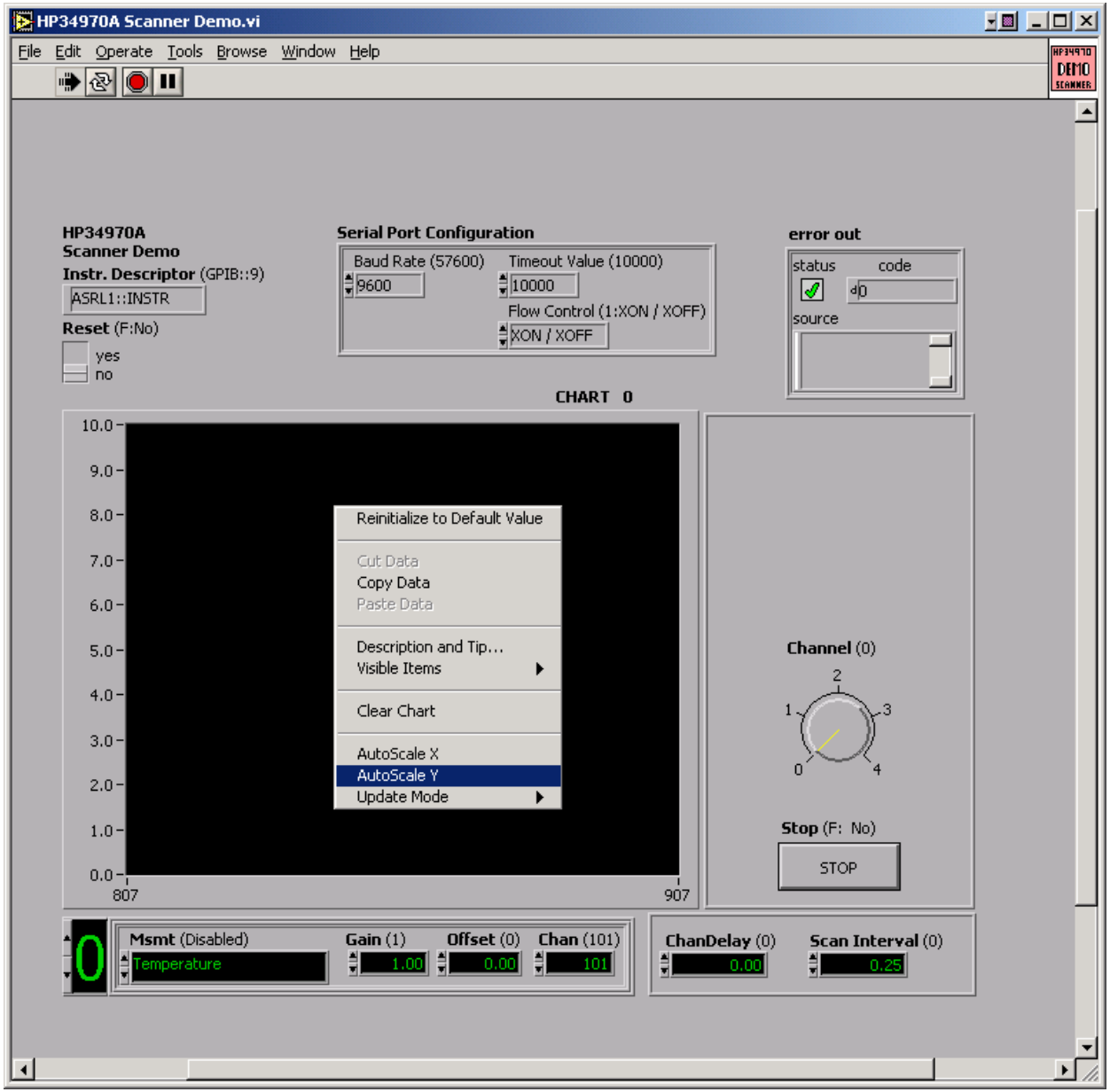
You'll see nothing at first. The DAQ unit should now be displaying "Scanning" and have an asterisk flashing.

You'll also notice that the arrow and the nearby menu buttons have changed…

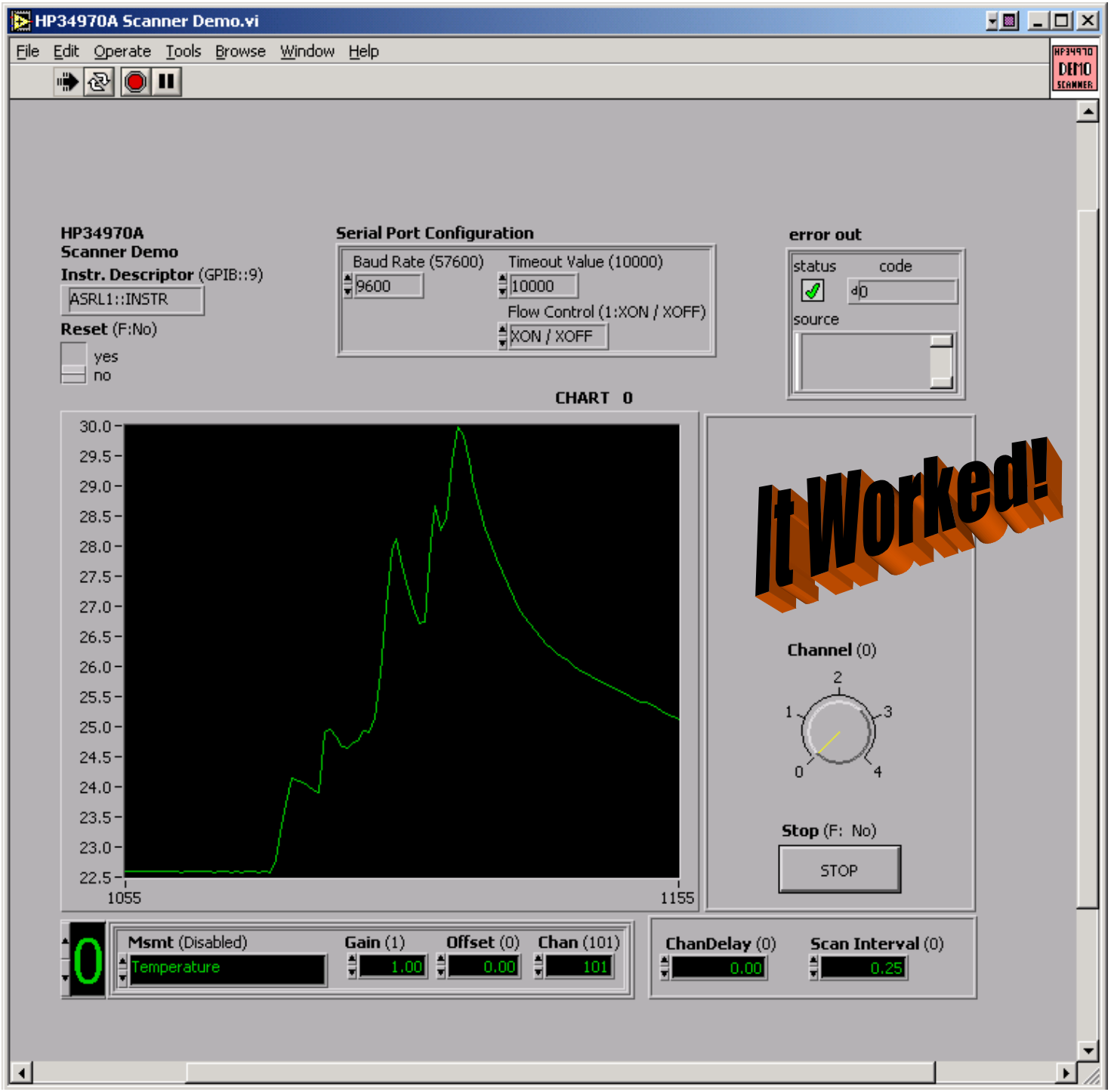
Why don't we see anything???

- We need to change the limits of the display! The chart is displaying values between 0 and 10, but our expected Temperature is in the 20 neighborhood.

- You may have noticed that the x-axis labels are changing…

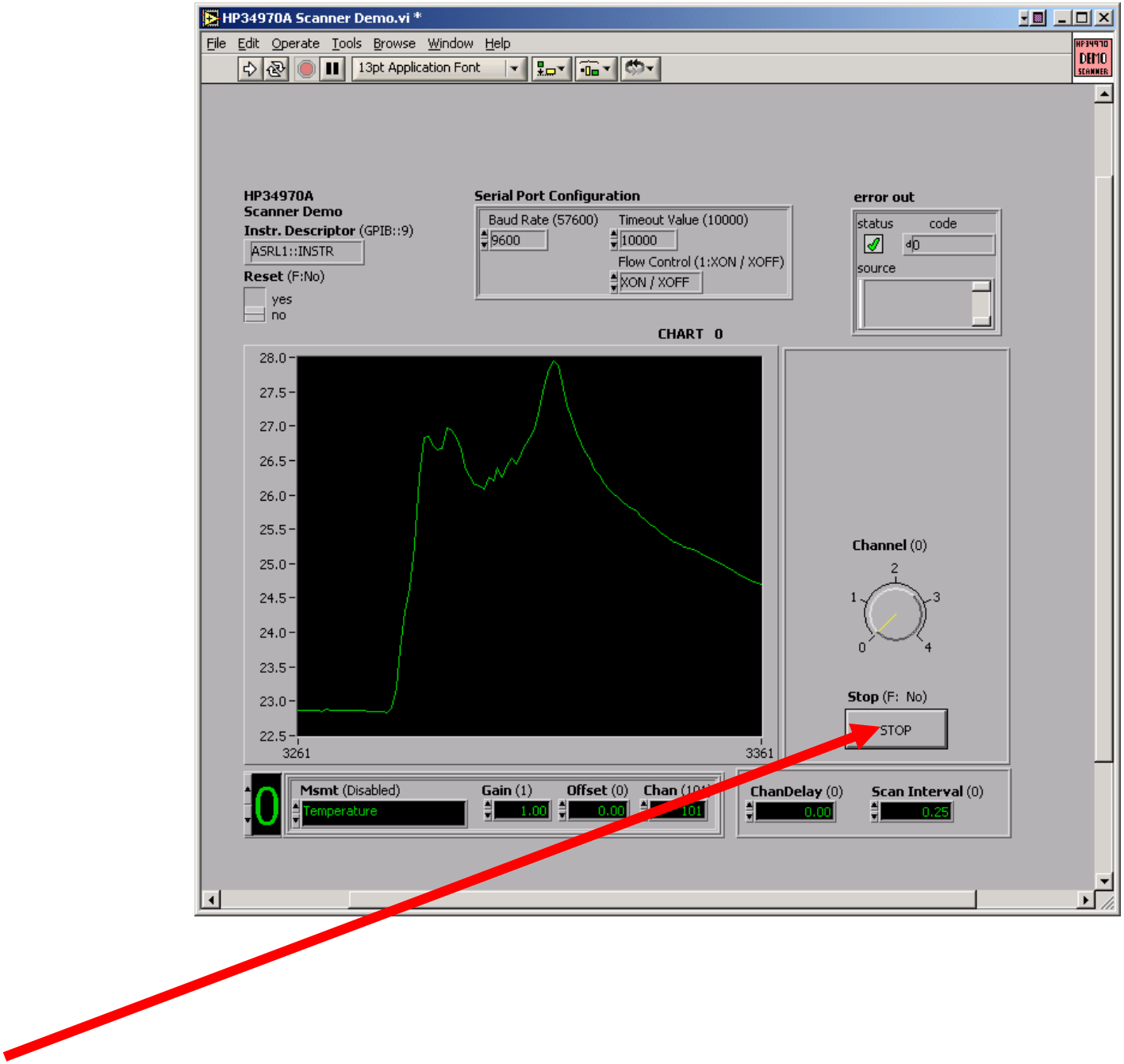


Right click in the chart area and select the Autoscale Y value…


HP34970A Scanner Demo.vi
File Edit Operate Tools Browse Window Help
HP34970A
Scanner Demo
Instr. Descriptor (GPIB::9)
ASRL1::INSTR
Reset (F:No)
yes
no
Serial Port Configuration
Baud Rate (57600)
9600
Timeout Value (10000)
10000
Flow Control (1:XON / XOFF)
XON / XOFF
error out
status
code
0
source
CHART 0
30.0
29.5
29.0
28.5
28.0
27.5
27.0
26.5
26.0
25.5
25.0
24.5
24.0
23.5
23.0
22.5
1055
1155
It Worked!
Channel (0)
2
1
3
0
4
Stop (F: No)
STOP
0
Msmt (Disabled)
Temperature
Gain (1)
1.00
Offset (0)
0.00
Chan (101)
101
ChanDelay (0)
0.00
Scan Interval (0)
0.25

- By the way, screen shots are made by selecting Alt-Print Screen, which copies the contents of the active window onto the clipboard. You can then paste these to MS Word for your documentation.

- You should probably select the Stop button on the Scanner Demo VI…


HP34970A Scanner Demo.vi *
File Edit Operate Tools Browse Window Help
13pt Application Font
HP34970A
Scanner Demo
Instr. Descriptor (GPIB::9)
ASRL1::INSTR
Reset (F:No)
yes
no
Serial Port Configuration
Baud Rate (57600)
9600
Timeout Value (10000)
10000
Flow Control (1:XON / XOFF)
XON / XOFF
error out
status
code
source
CHART 0
28.0
27.5
27.0
26.5
26.0
25.5
25.0
24.5
24.0
23.5
23.0
22.5
3261
3361
Channel (0)
Stop (F: No)
STOP
Msmt (Disabled)
Temperature
Gain (1)
1.00
Offset (0)
0.00
ChanDelay (0)
0.00
Scan Interval (0)
0.25

- Now it’s ready to run again if you would like to gain more experience with this demo. Be sure to look at the Diagram window for this VI and complete your laboratory tasks…